An Efficient FPGA Implementation of Spiking Neural Networks for N-MNIST Classification

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Abstract

- Implemented Spiking Neural Network (SNN) for N-MNIST image classification in Artix-7 FPGA (Basys 3).
  - High Accuracy : 96.5%
  - Low Resource Utilization : 4,808 LUTs and 2 DSPs

Outline

- Introduction
- Literature Review
- Proposed Methodology
- Results and Discussion
- Conclusion

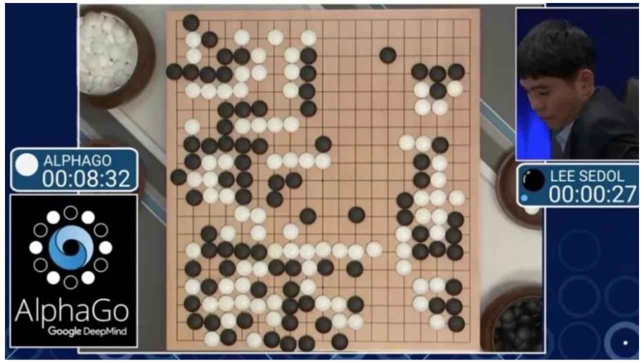
Introduction

Why SNN? **Why LIF Neuron?** **Why FPGA Implementation?**

Introduction

Why SNN? Why LIF Neuron? Why FPGA Implementation?

AI now



AlphaGo
Recognition 250W

Smart Glass
2.1Wh drains 25 mins

Autonomous cars
383 GB to 5.17 TB an hour

62,318 MWh
6.9K tons CO_2 emission

Roy, Kaushik, Akhilesh Jaiswal, and Priyadarshini Panda. "Towards spike-based machine intelligence with neuromorphic computing." Nature 575.7784 (2019): 607-617

Introduction

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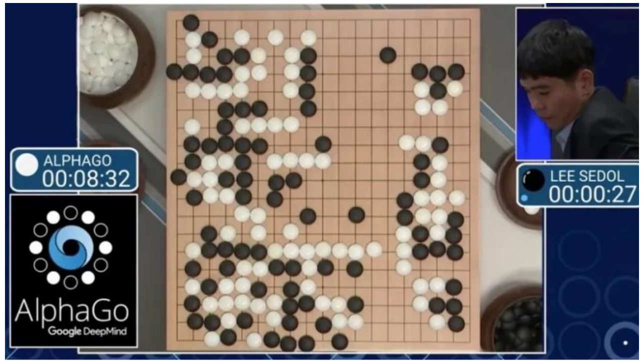

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Human Brain: 20W

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- **Biological realism**: Mimic actual neuron communication via discrete spiking neuron
- **Energy efficiency**: Event-driven computation, sparse activation
- **Hardware advantage**: Binary spike signals reduce data movement & power
- **Edge AI**: 100-1000 × lower power than ANNs for same accuracy

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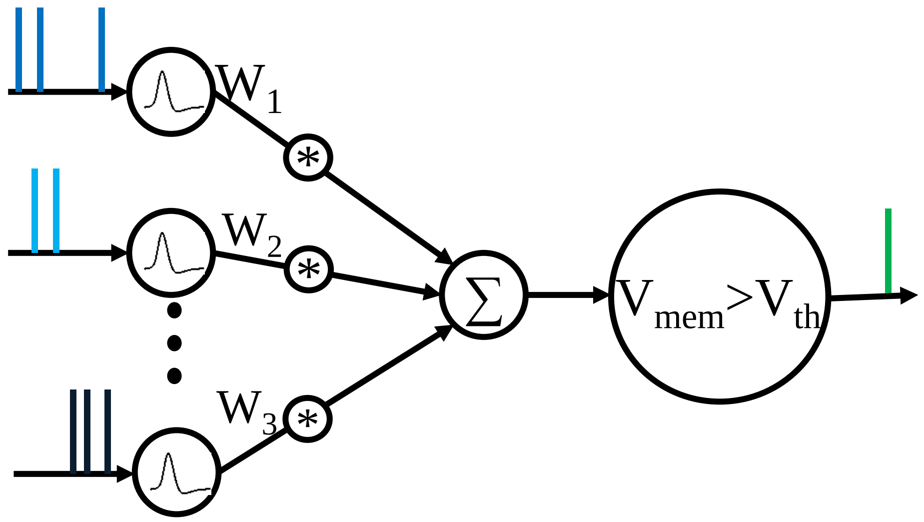
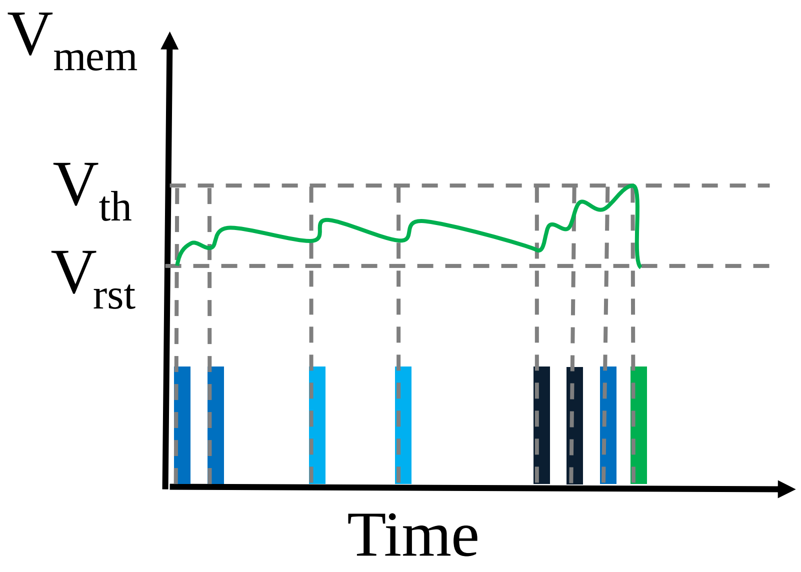
Introduction

Why SNN? **Why LIF Neuron?** Why FPGA Implementation?

Models for Spiking Neuron		
Model	**No. of Variable**	**Complexity**
Leaky Integrate & Fire (LIF)	1	Very Low
Izhikevich	2	Very Low
Hodgkin Huxley	4	Very High





Tan, Clarence, Marko Šarlija, and Nikola Kasabov. "Spiking neural networks: background, recent development and the NeuCube architecture." Neural Processing Letters 52 (2020): 1675-1701.

Introduction

Why SNN? **Why LIF Neuron?** Why FPGA Implementation?

- **Balance**: Simple yet biologically plausible temporal dynamics
- **Hardware friendly**: Requires only accumulation, leak, threshold reset
- **Proven accuracy**: Excellent recognition performance on standard benchmarks
- **Scalability**: Easy to implement on resource constrained FPGAs

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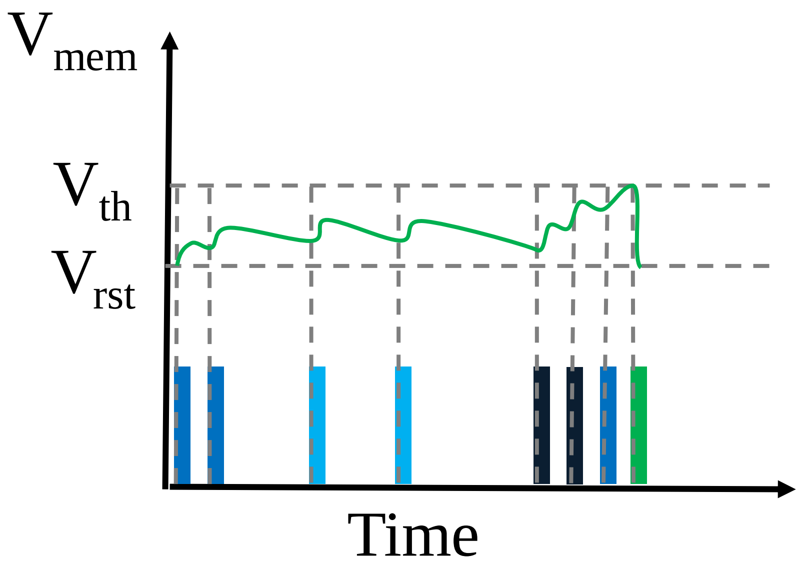

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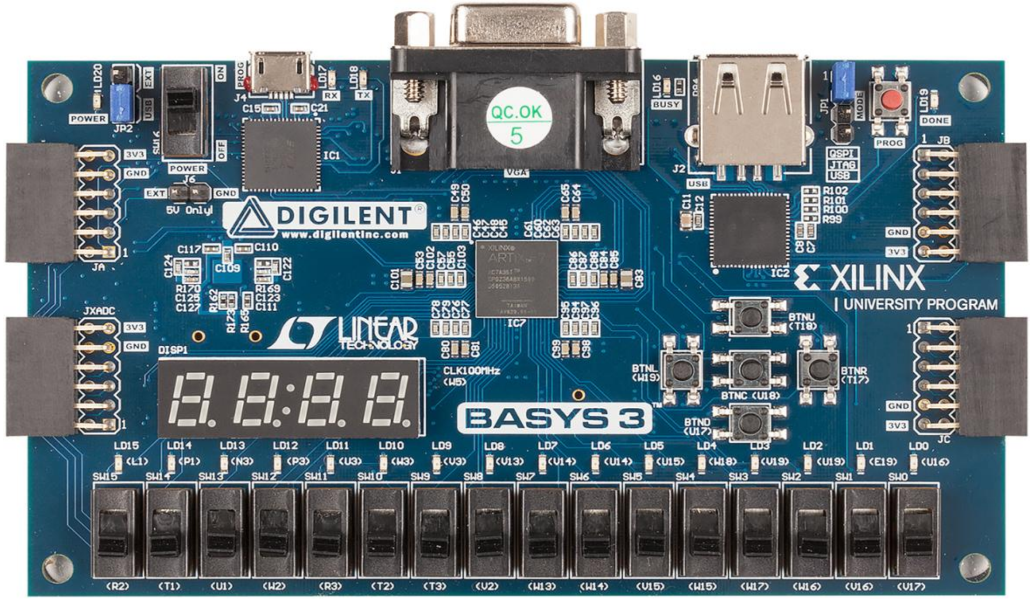
Introduction

Why SNN? | Why LIF Neuron? | **Why FPGA Implementation?**

- **Reconfigurability**: Change network instantly without recompiling
- **Prototyping:** fast validation of novel architectures (like our Bias-Free Neuron) before ASIC fabrication.
- **Real-time**: Deterministic latency, no OS jitter for edge devices
- **Cost-effective**: 100-1000× cheaper than ASICs, accessible to researchers



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State-of-the-Art

Mishra et al. (2025) - Adaptive LIF on Artix-7:
95% MNIST accuracy | 79,775 LUTs | 0.9ms latency | 426mW | Extreme resource demand

Carpegna et al. (2022) - Spiker Accelerator:
74% MNIST accuracy | 29,145 LUTs | 215µs latency | 6W power | Low accuracy, high power

Ma et al. (2017) - Darwin NPU (Spartan-6):
92% accuracy | 27,288 LUTs | 152ms latency | 1.5W power | High power consumption

Vipinkmenon (2020) - Standard ANN (Artix-7):
95% accuracy | 6,895 LUTs | 160 DSPs | 206W | DSP Intensive (160 vs 2) & High Power

- Massive LUT consumption limits low-end FPGA deployment.
- Accuracy-efficiency trade-off: high accuracy requires high resources.
- Power consumption too high for edge devices.

Proposed Methodology

SNN Implementation
Three layer SNN 784-100-10

Portable
Power utilization Resource utilization

Low Level Board
Basys-3 (Artix-7 xc7a35t)

Accuracy
High

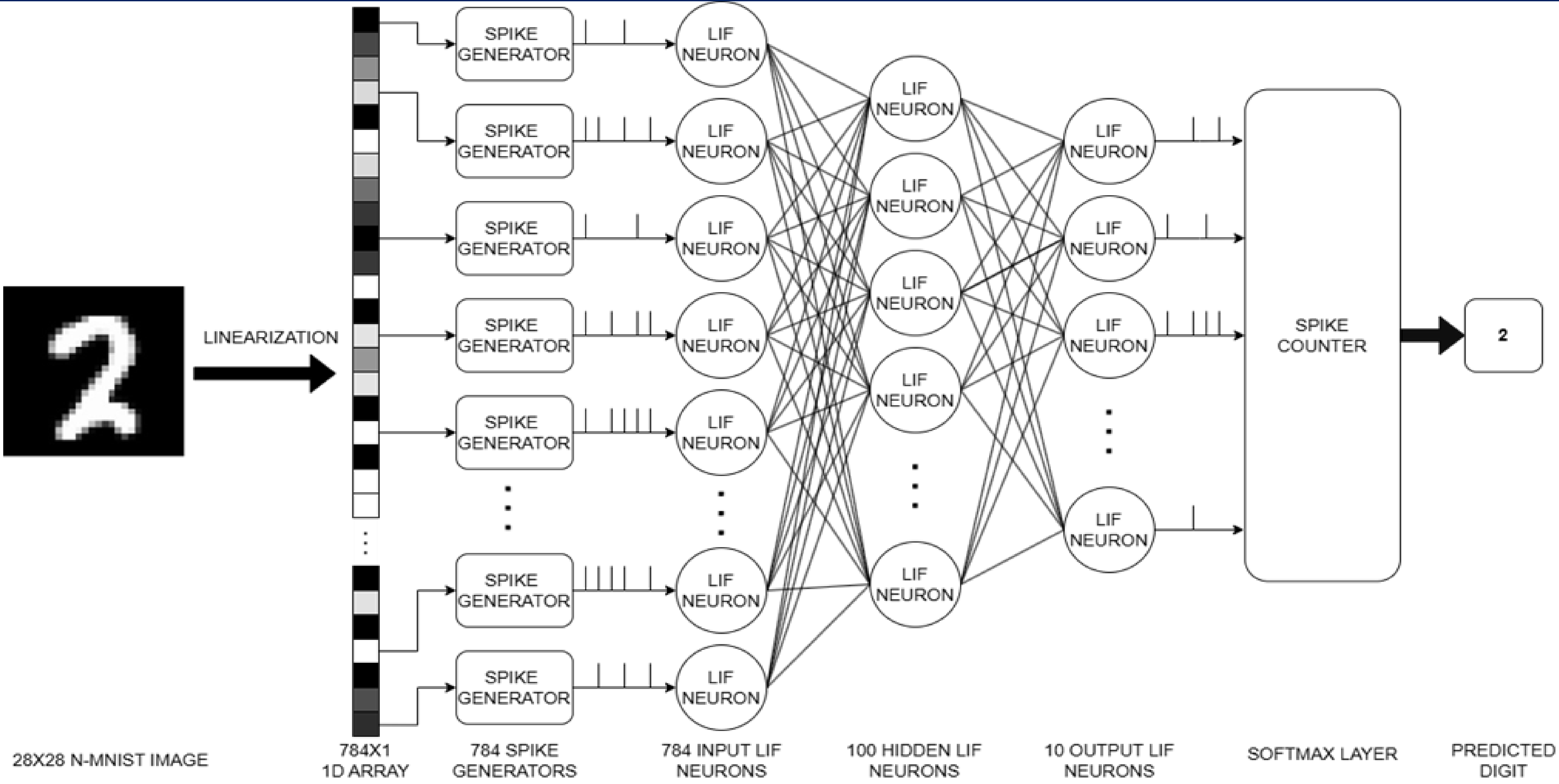
- **Bias-free digital LIF neurons**
- Multiplier-less synaptic operations
- **Serial time-multiplexed dataflow**
- Fully on-chip storage using BRAM
- Hardware aware off chip training in Python.

Network Parameters

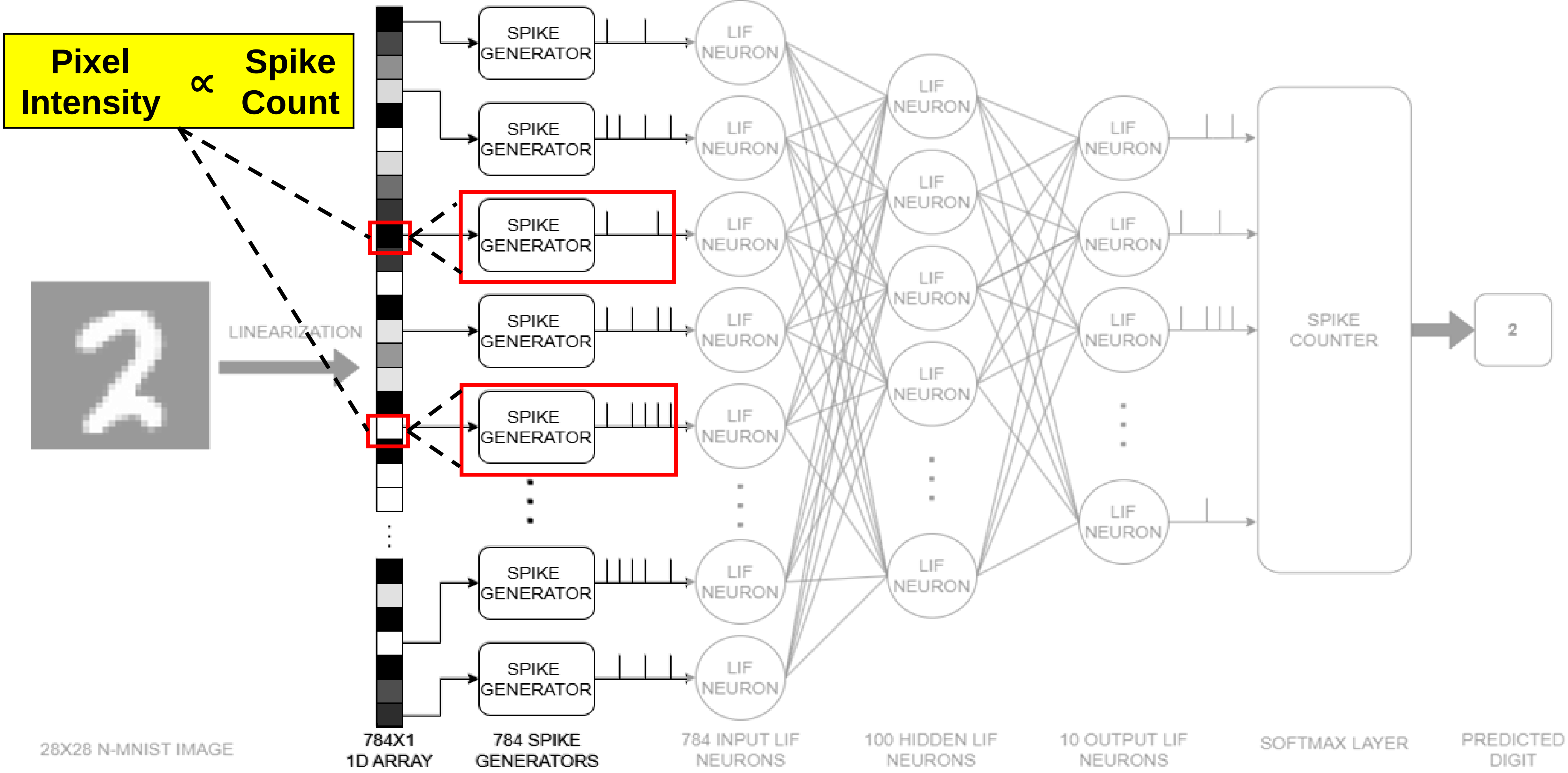
Neuron	Synapse	Spike Generation
• 884 Neurons • LIF Neuron	• 79400 • 77.354 kB	• Rate coding • 25 steps

Clock Frequency	100 MHz

FPGA Implementation Flow of the SNN



FPGA Implementation Flow of the SNN



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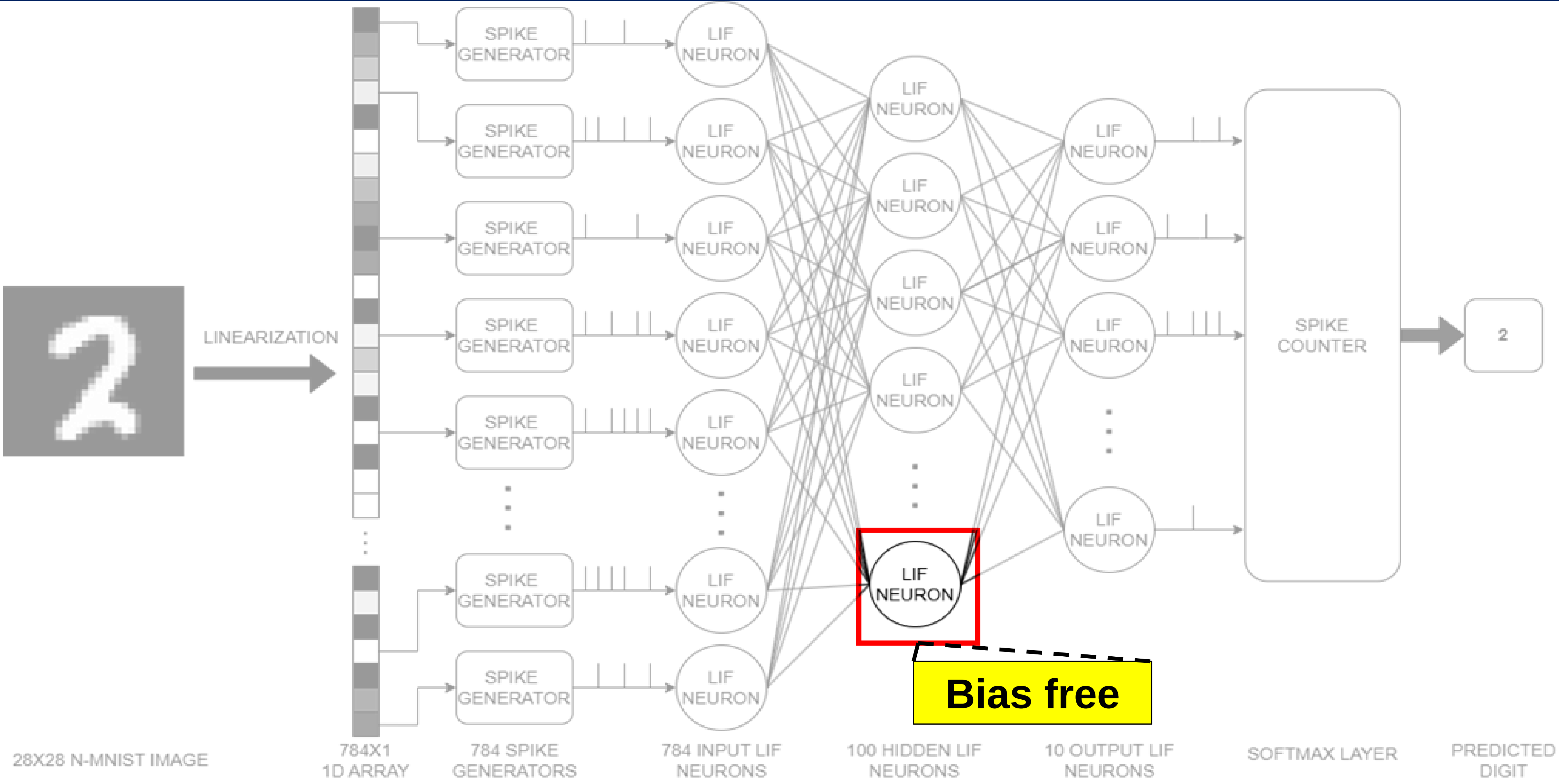

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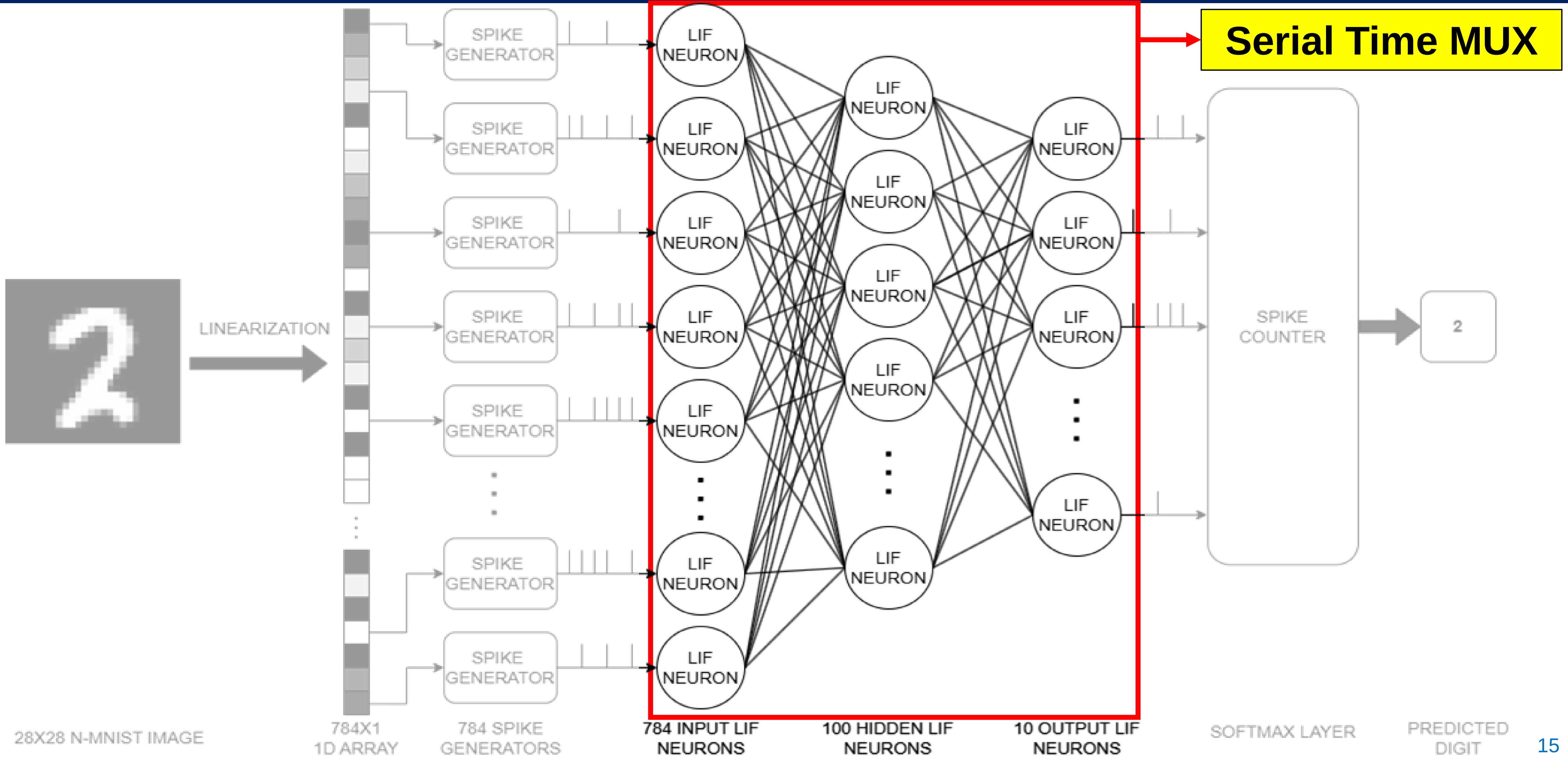


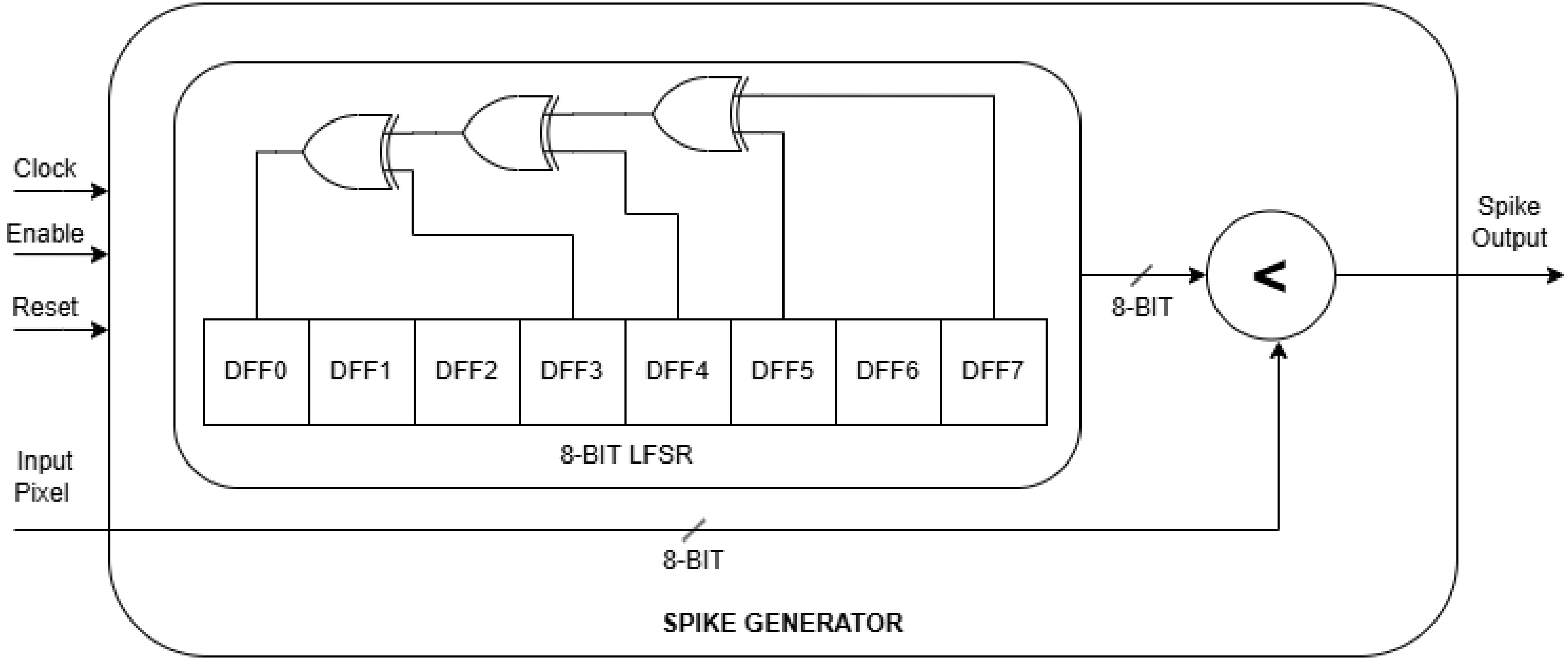
FPGA Implementation Flow of the SNN

Spike Generator	Digital LIF Neuron	Serial Time Multiplexer

FPGA Implementation Flow of the SNN

Spike Generator	Digital LIF Neuron	Serial Time Multiplexer

- 8-bit Linear Feedback Shift Register (LFSR) to generate pseudo-random values
- Spike is produced when LFSR value < Pixel Value



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FPGA Implementation Flow of the SNN

Spike Generator	Digital LIF Neuron	Serial Time Multiplexer

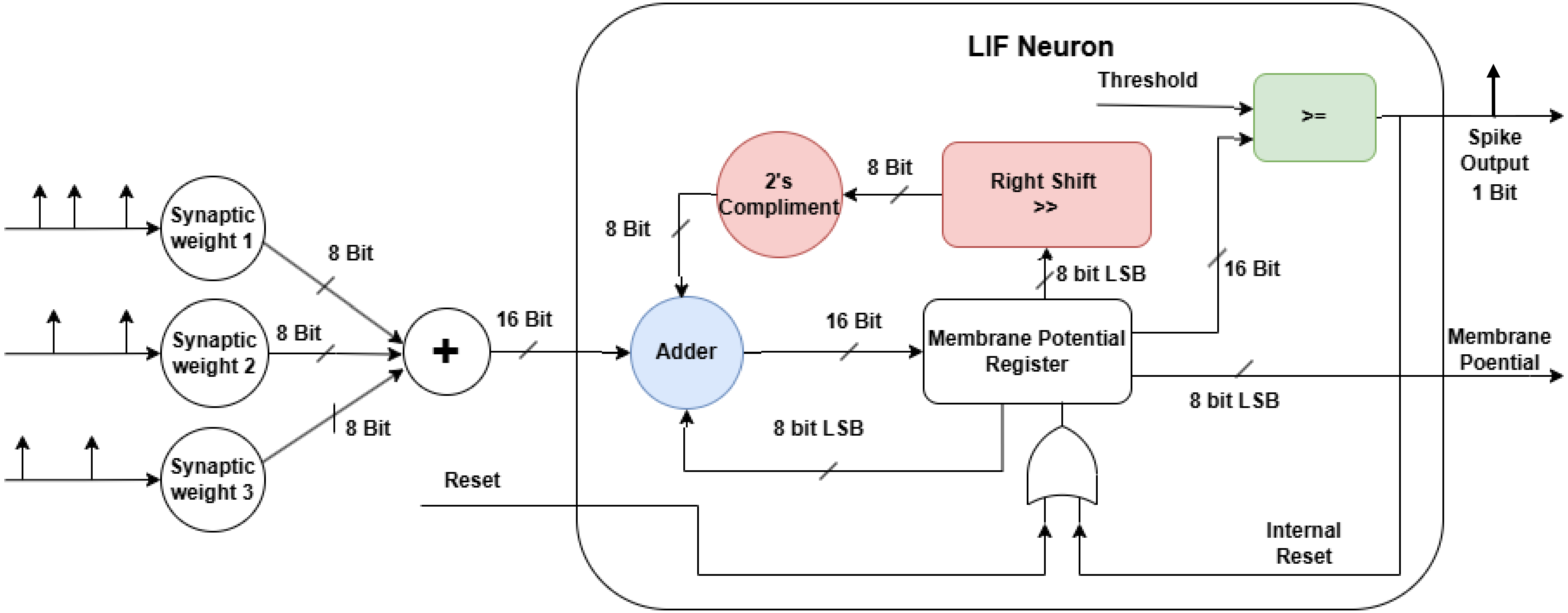
Traditional equation : $V(t+1) = \alpha . V(t) + \sum W . S(t) + Bias$

Bias-free equation : $V(t+1) = \alpha . V(t) + \sum W . S(t)$

Symbol	Parameter	Bit Length	Range
$V(t)$	Membrane Potential at time 't'	16 Bit	0 to 127
$S(t)$	Spike at time t	1 Bit	0 or 1
W	Weights	8 Bit	-128 to 127
α	Decay factor	>> 4	0.75 to 1
$Bias$	Threshold offset	-	-

FPGA Implementation Flow of the SNN

Spike Generator	Digital LIF Neuron	Serial Time Multiplexer



Leaky Integrate Fire

FPGA Implementation Flow of the SNN

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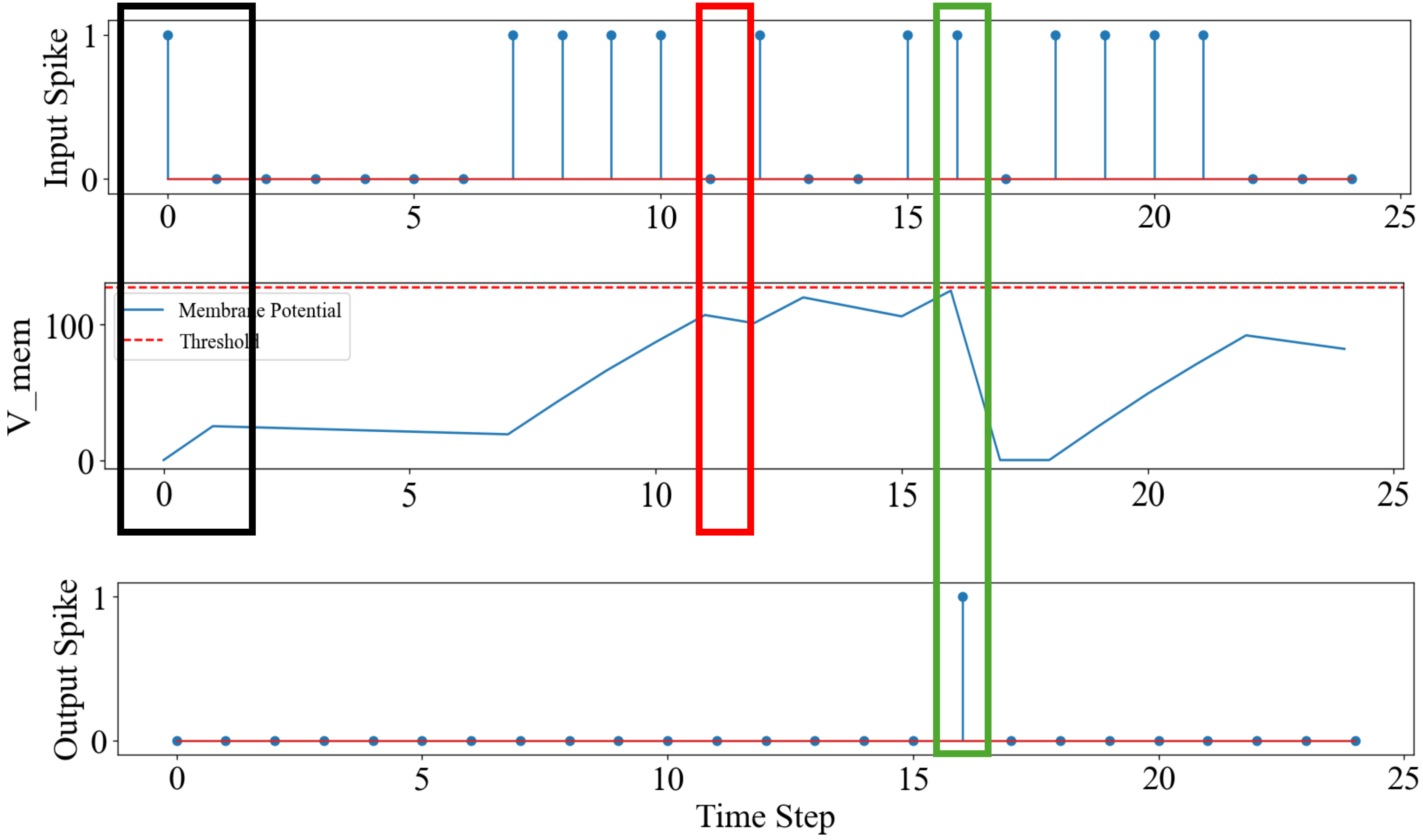

Leak

Integrate

Fire

Simulated in Python

FPGA Implementation Flow of the SNN

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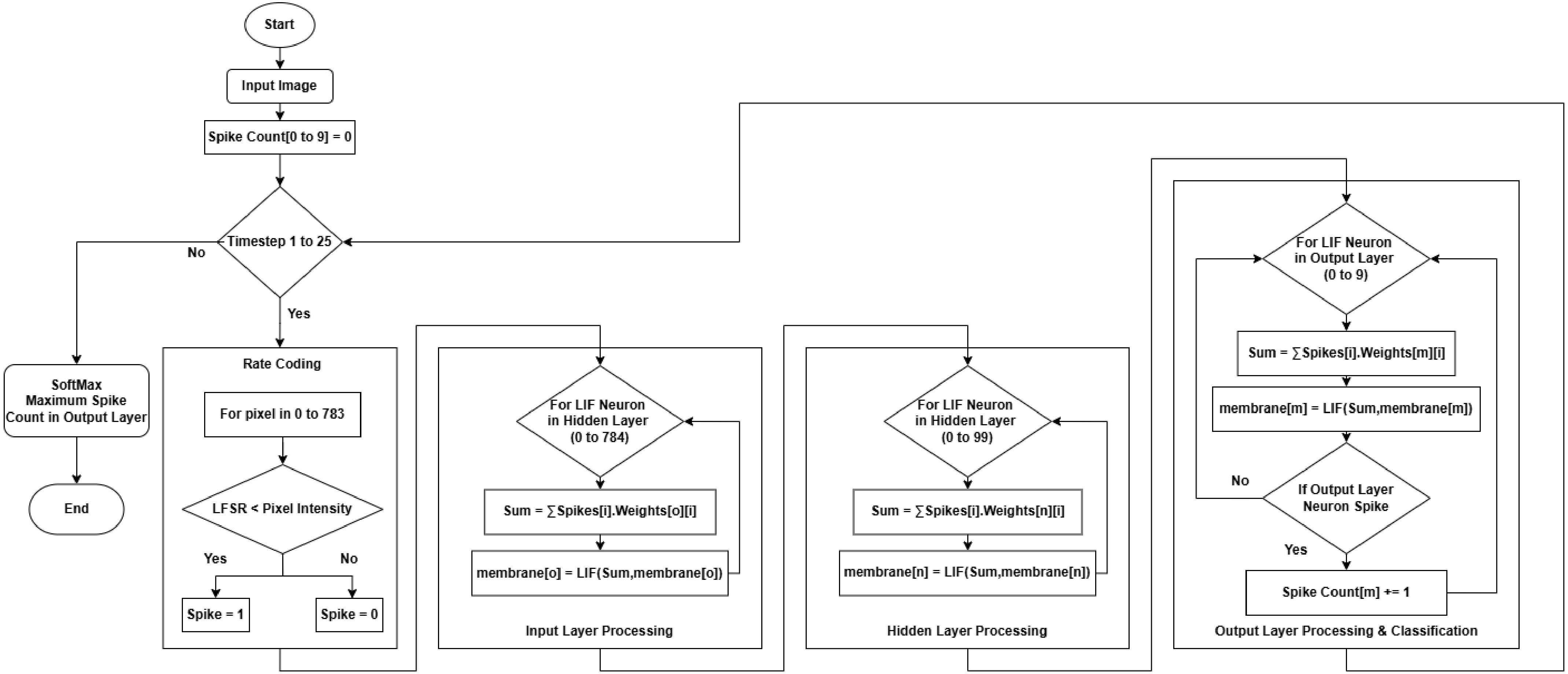
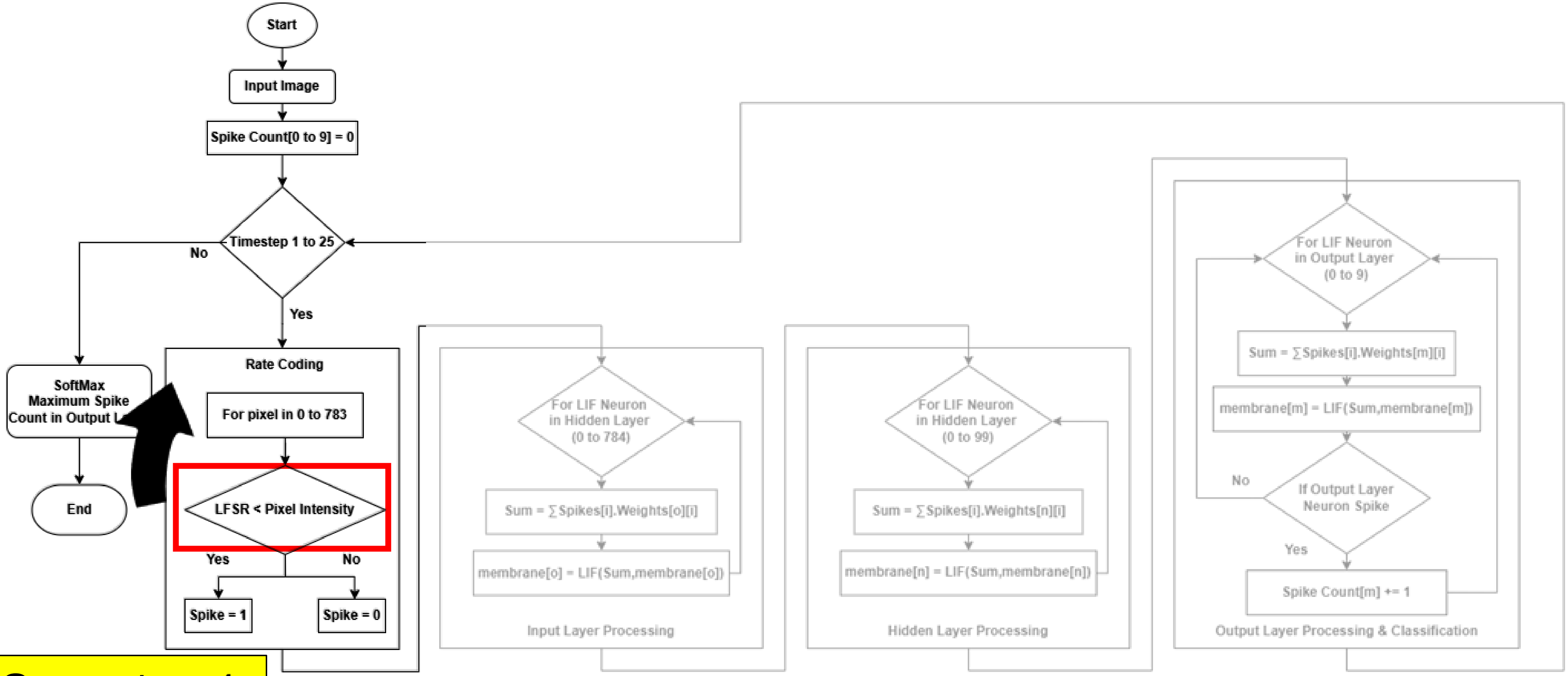


FPGA Implementation Flow of the SNN

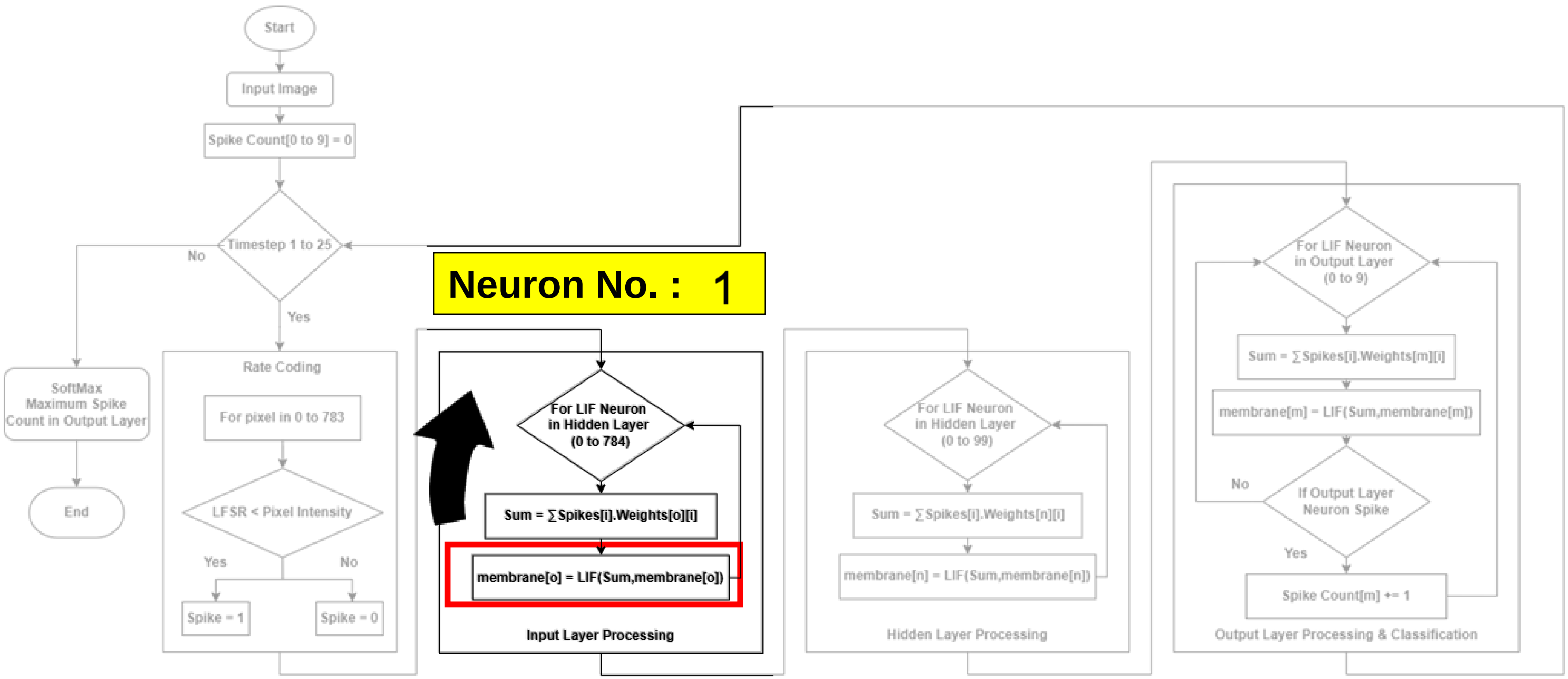
Spike Generator	Digital LIF Neuron	Serial Time Multiplexer



Spike Generator: 1

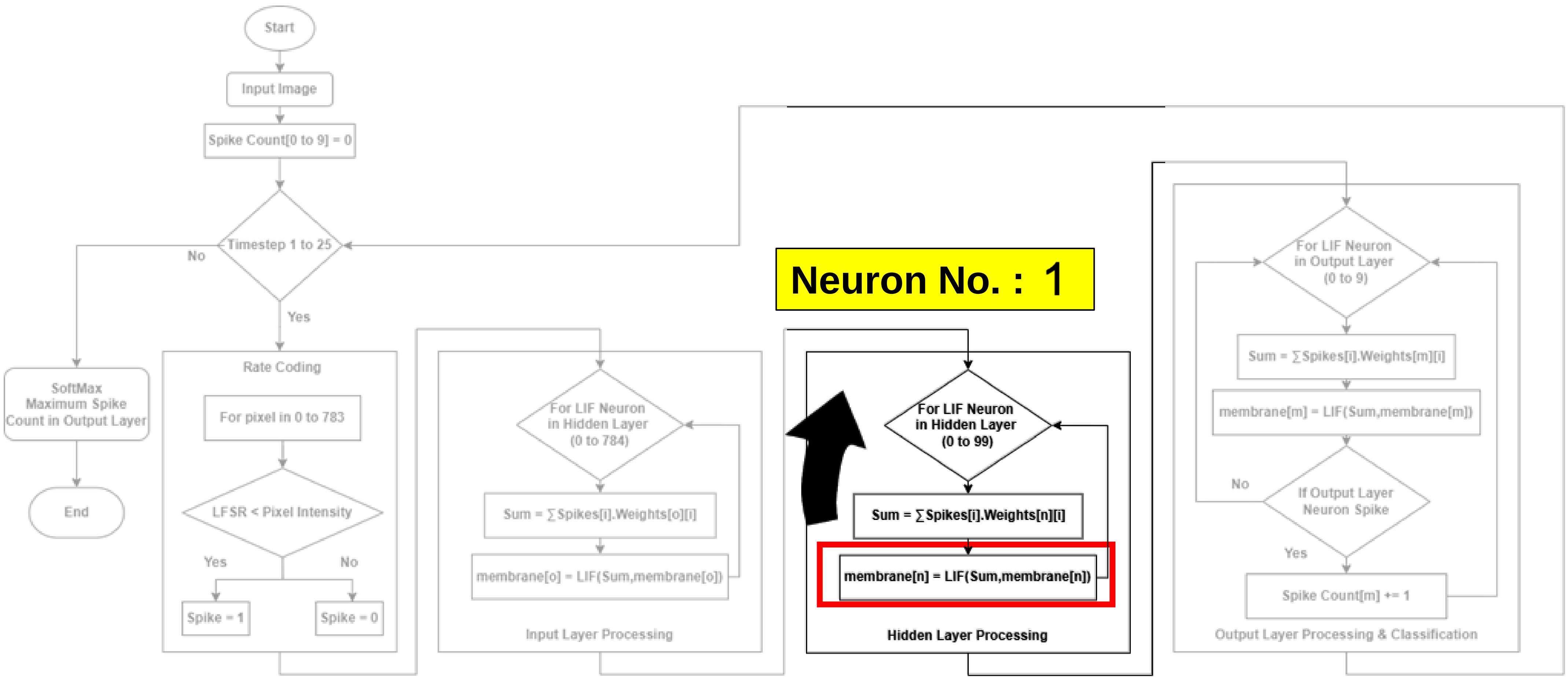
FPGA Implementation Flow of the SNN

Spike Generator	Digital LIF Neuron	Serial Time Multiplexer



FPGA Implementation Flow of the SNN

Spike Generator	Digital LIF Neuron	Serial Time Multiplexer



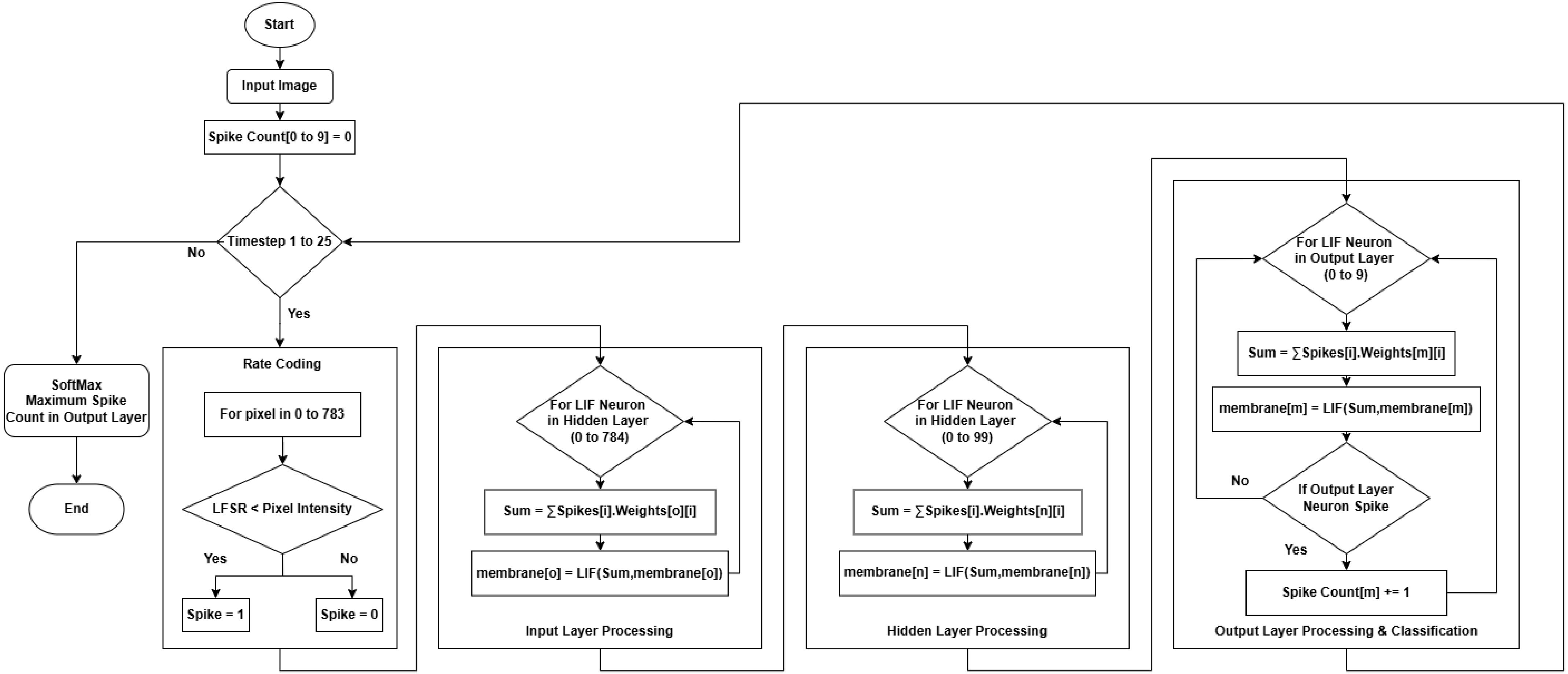
FPGA Implementation Flow of the SNN

Spike Generator	Digital LIF Neuron	Serial Time Multiplexer

Neuron No. : 1

FPGA Implementation Flow of the SNN

Spike Generator	Digital LIF Neuron	Serial Time Multiplexer



FPGA Implementation Flow of the SNN

Spike Generator	Digital LIF Neuron	**Serial Time Multiplexer**

- Reuses a single LIF Neuron for computation
- Processes neurons and layers sequentially
- Drastically reduces LUT and DSP usage
- Trade-off: Exchanges extreme parallel speed for maximum hardware efficiency.

Experimental Results

- Achieved **96.5%** test accuracy on MNIST
- No accuracy drop between Python simulation and FPGA execution
- Inference latency: **4.4 ms per image** (at 100 MHz)
- LUT count **4808**
- On-chip power: **150mW –190 mW**

FPGA Resource Usage

Artix-7 (xc7a35t) Utilization

Resource	Used	Available	Utilization (%)
Slice LUTs	4808	20800	23.1 %
Slice Registers (FF)	4476	41600	10.8 %
F7 Muxes	33	16300	0.20 %
Block RAM	24.5	50	49 %
DSP Slices	2	90	2.2 %
Carry Locgic (CARRY4)	697	5200	13.4 %
Global Buffers (BUFG)	1	32	3.1 %

Results: Comparison with State-of-the-Art

	This work	[1]	[2]	[3]	[4]	[5] ANN
Slice LUT	4808	27288	79,775	29145	22000	6895
Architecture	784:100:10	784:500:500:10	784:1000:10	784:500:500:10	784:500:500:10	784:30:30:10:10
BRAM	25	116	107	45	200	15
DSP	2	58	0	-	-	160
Power	190mW	1.5W	426mW	6W	1.5W	206W
Accuracy	96%	92	95%	74%	92	95%
Computation time	4.4ms	152ms	0.9ms	215ms	236us	-
Neuron model	LIF	LIF	Adaptive LIF	LIF	LIF	RELU
Synapse	79100	647000	647,000	313600	647000	24870
Clock frequency	100MHz	75MHz	100MHz	100MHz	75MHz	100MHz
Platform	Artix-7	Spartan 6	Artix-7	Artix-7	Spartan 6	Artix-7

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[1] D. Ma, J. Shen, Z. Gu, M. Zhang, X. Zhu, X. Xu, Q. Xu, Y. Shen, and G. Pan, “Darwin: A neuromorphic hardware co-processor based on spiking neural networks,” Journal of systems architecture, vol. 77, pp. 43–51, 2017.

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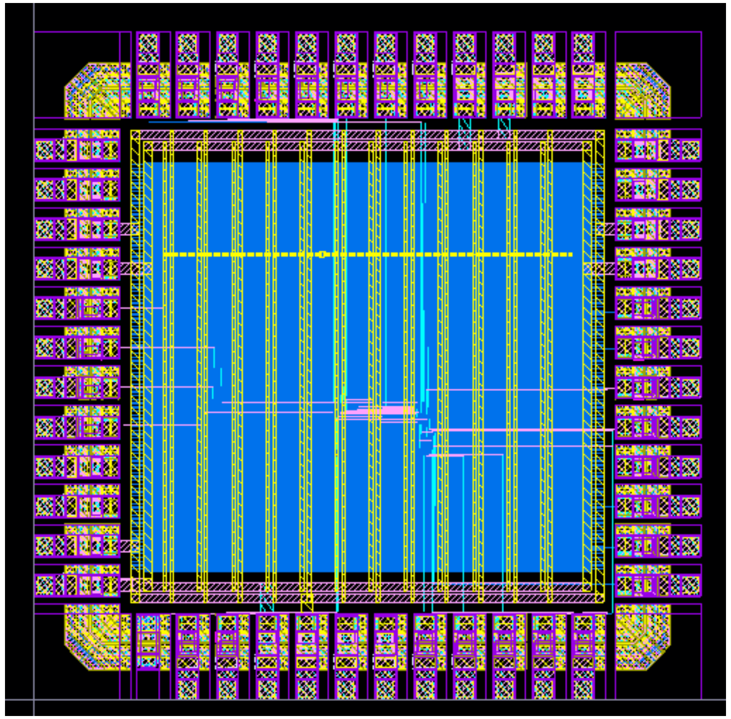
Future Works

- Tape-out of this Spiking Neural Network using SCL 180nm PDK
- Enabling lightweight STDP algorithm for on-chip learning



Implemented Digital LIF Neuron in SCL 180nm PDK

Conclusion

Bias-free LIF offer better trade-off between power and delay in achieving better classification accuracy at low-end FPGA board.

- Accuracy : 96.5% on the MNIST benchmark.
- Low Resource Utilization : 4,808 LUTs and 2 DSP slices.
- On-chip power : 150mW –190 mW
- Delay : 4.4ms

References

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